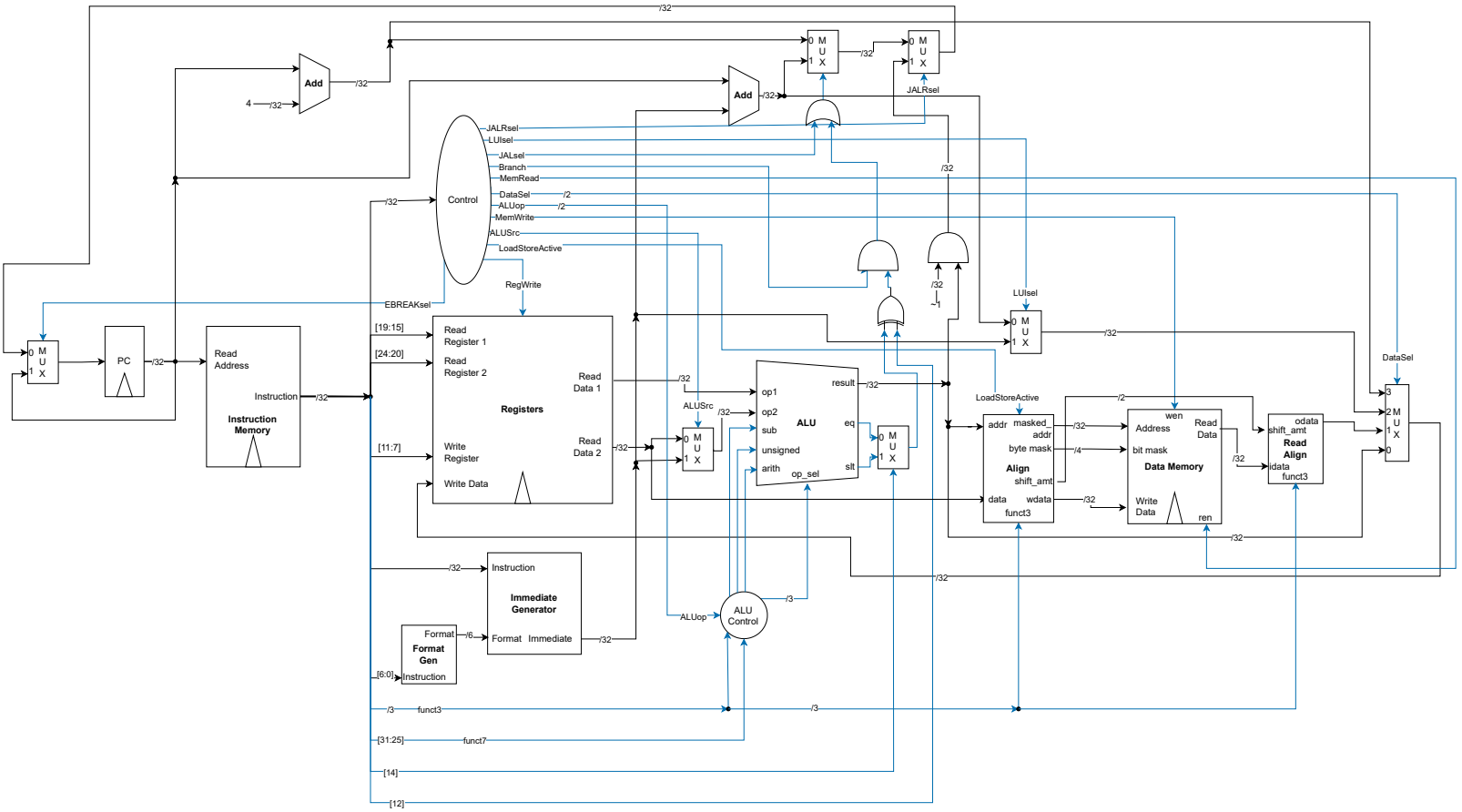

/32
Add
4
/32
/32
Add
/32
0 M U 1 X
/32
0 M U 1 X
JALRsel
JALRsel
LUIsel
JALsel
Branch
MemRead
DataSel
/2
ALUop
/2
MemWrite
ALUSrc
LoadStoreActive
Control
/32
RegWrite
EBREAKsel
/32
/32
1
LUIsel
0 M U 1 X
/32
DataSel
3
2 M U 1 X
0
0 M U 1 X
PC
/32
Read Address
Instruction
/32
Instruction Memory
[19:15]
[24:20]
[11:7]
Read Register 1
Read Register 2
Registers
Read Data 1
Read Data 2
Write Register
Write Data
/32
/32
ALUSrc
/32
0 M U 1 X
op1
op2
sub
unsigned
arith
ALU
result
eq
slt
op_sel
0 M U 1 X
/32
LoadStoreActive
/2
addr
masked addr
byte mask
Align
shift_amt
data
wdata
funct3
/32
/4
/32
wen
Address
Read Data
bit mask
Data Memory
Write Data
ren
/32
odata
shift_amt
Read Align
idata
funct3
/32
/32
/32
Instruction
Immediate Generator
Format
Immediate
/32
/32
ALUop
ALU Control
/3
Format
/6
Format Gen
Instruction
[6:0]
/3
funct3
/3
[31:25]
funct7
[14]
[12]